

Engineering

Henry Yeonjun Kim



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2026-01-21: First Tutorial – Key Contacts, Project Setup, and Initial Client-Meeting Scheduling

Team: 13

Team Directory

Name	Role / Program Note	Contact
Henry Kim	Quality Assurance Manager, TrackOne	
Merrick Wang	Team Leader, Chemical Engineering	
Omo Faseemo	Communications Manager, Chemical Engineering	
Shannon Ho	Project Manager, Chemical Engineering	
Kamilia Desroches Parchment	Meetings coordinator, Chemical Engineering	
Abdulrahman Abuloghod	Research Coordinator, Chemical Engineering	

Client and Course Instructors Contacts

Contact	Recorded information	Contact details
Client	Alexandre Klaus, Skule Kup Director	
Teaching Assistant (TA)	Ash Mahmood (Masters in bio-eng)	preferred: kash- secondary:
Communication Instructor (CI)	Ebru Bag	
Engineering Manager (EM)	Glenn Hibbard (Prof at U of T)	preferred: secondary:



Project Brief



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

Engineering Strategies & Practice

APS112 - Winter 2026

Project Description

Project/Team:	13
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CLIENT INFORMATION

Client Name(s):	Alexandre Klaus
Professional Title, Occupation, or Identifier in relation to the project:	Skule Kup Director
Company:	
Address:	
Email:	kup@skule.ca
Telephone:	6479743544
Alternate Contact Name:	
Alternate Contact Email:	
Alternate Contact Telephone:	
Project Title:	Buzzer beaters: A portable buzzer system.
Need to develop a compact and portable buzzer system that indicates first responder in a jeopardy style. Should be able to have up to 8 buzzers connected, and as few as two. This would be used in certain Skule Kup events to allow for a better format to things like Discipline Feud, the typical Skule Kup closing event.	

Weekly Team Meeting Setup

- Weekly team meeting: Tuesdays, 6:00–7:00 PM, Gerstein Library.
- Room booking: Kamilia.
- Virtual attendance: used if needed.

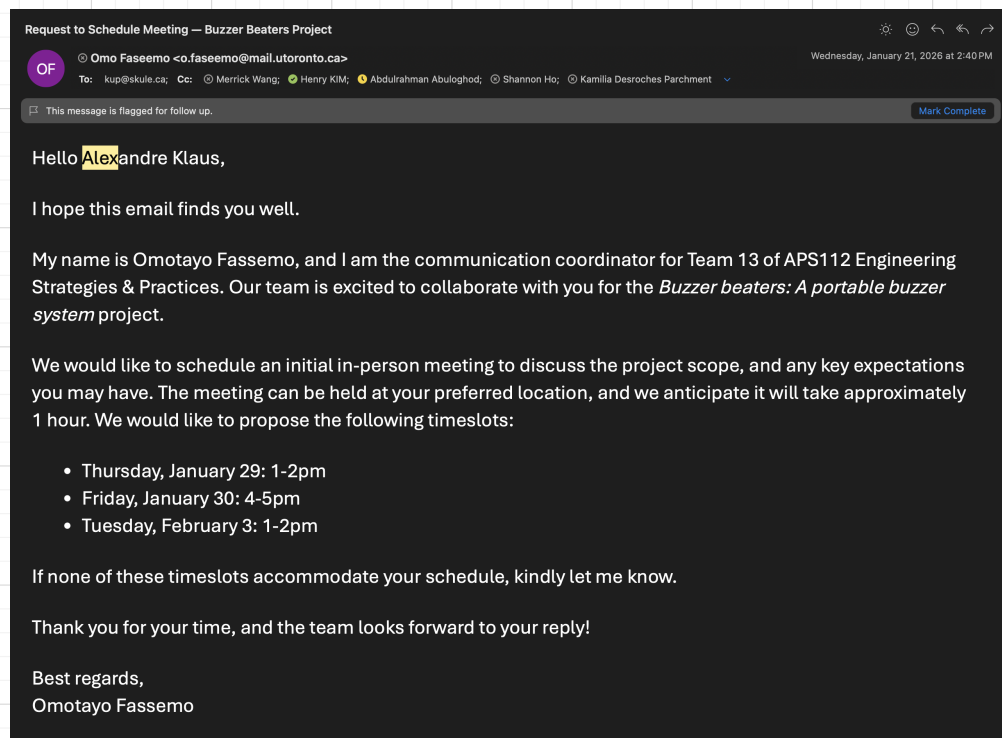


First Client Meeting Candidate Times

- January 29, 2026 from 1:00–2:00 PM
- February 3, 2026 from 1:00–2:00 PM
- January 30, 2026 from 4:00–5:00 PM

Team Formation Notes

- The tutorial agenda included forming into teams, starting the **Team Charter**, receiving the project, and writing a **Client Email**.
- Shared team-member names and contact information.
- Recorded client, TA, CI, and EM contact information.
- The client email was prepared, approved by the **EM**, and then sent that day to arrange the first client meeting.
- The approved email proposed three meeting times: **January 29, 1:00–2:00 PM, January 30, 4:00–5:00 PM, and February 3, 1:00–2:00 PM.**
- Proposed candidate times for the first client meeting.





Week 3 To-Do List

- Team charter needed to be completed by the Wednesday before tutorial; this was done on January 27, 2026.
- PLR review was due the following week on February 1, 2026 at 1:00 PM.

2026-01-27: First Team Meeting

- First in-person team meeting, held in B105E at Gerstein Library from 6:00–7:00 PM.
- Main focus: finalizing the team charter and preparing for the first client meeting.
- The team finalized the team charter during this session.
- The team began discussing the upcoming client meeting.
- The team reviewed how to improve one another's team charter sections.

2026-01-28: Second Tutorial

- The tutorial focused on preparing the **Client Meeting Plan**.
- During the tutorial, the team discussed what questions to ask in the **first client meeting** scheduled for the next day.
- This tutorial also fit into the course requirement that the client meeting plan needed **EM approval** before meeting the client.

2026-01-29: First Client Meeting - Galbraith Lobby @ 1PM

- Top priority: keep the buzzer **compact and portable**.
- Size target recorded: about **30 × 30 × 30 cm**.
- **Usability** came before compactness and portability.
- Power source was discussed, but it was **not a top priority**.
- Budget target recorded: about **CAD 250**.
- The client slightly preferred a **lighting cue** over sound.
- Main event context: **Discipline Feud**.
- Runtime target: **more than two hours** if battery-powered.
- Existing commercial buzzers were considered too small or not scalable enough.



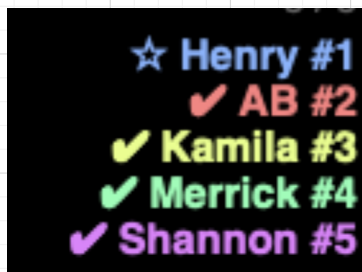
- Preferred form: **handheld buzzers**, about the weight of a water bottle.
- Preferred connection: **wired**, not Bluetooth.
- Required feature: **lockout system**.
- Durability note: survive about **20 cm** drop / table-height handling.
- Preferred materials: **MyFab-compatible**, especially **3D-printed parts** and **plywood**.
- Safety note: avoid **fire or shocking hazards**.
- Storage / environment: fit in a locker and aim for **IP66** water resistance.
- Recorded stakeholders: **Alexandre Klaus** and a future Skule Kup director.
- Event notes: up to **8 teams**, about **10–40 attendees**, indoor Pit setting.

Summary of Client Meeting 1

- Main priorities: **compactness**, **portability**, and **usability**.
- Main constraints: **2–8 players**, about **CAD 250**, **wired communication**, **lockout**, and about **two hours** of runtime.
- Other key notes: **handheld use**, **drop tolerance**, **locker storage**, **water resistance**, and **MyFab-compatible materials**.

2026-02-03: Second Team Meeting

- Second team meeting, focused on planning for the upcoming **Status Reports (SRs)**.
- **SR Part A** needed to be submitted **24 hours before the tutorial**.
- The SR required material related to the **Gantt chart**, **project manager**, and **two oral-presentation slides** on a rotating weekly basis.



SR notes.

- The team decided the upcoming **Status Report (SR)** order during this meeting.
- **Omo** volunteered to take the **first** slot, which is why his name does not appear in the image list.
- I later switched positions with **AB** because I had **two midterms in one week** during that period.

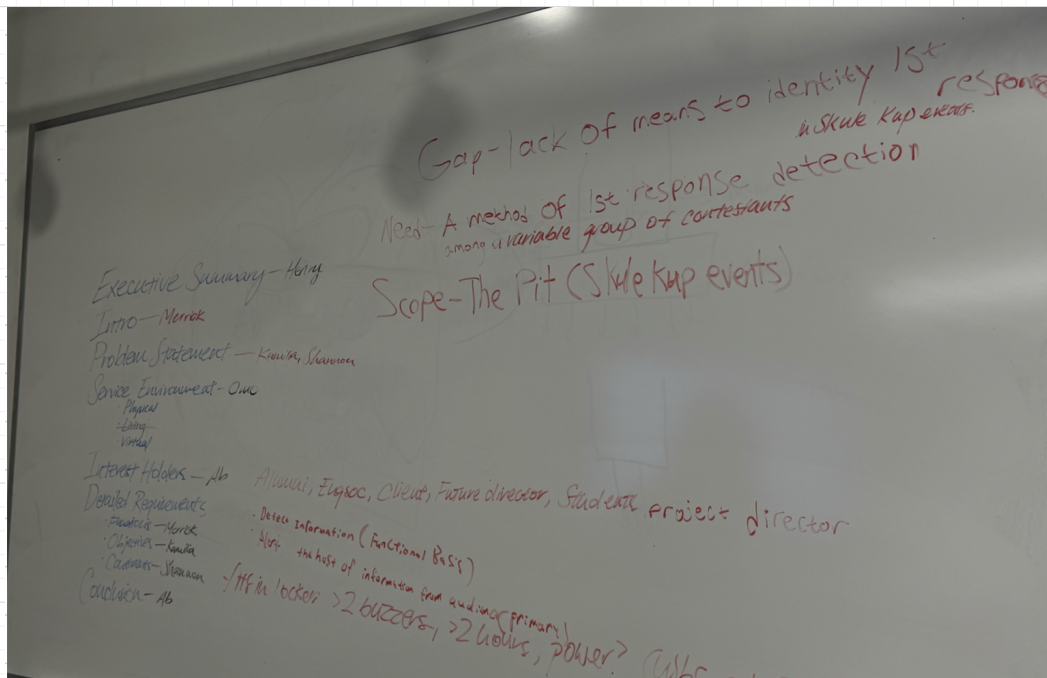


PR Task Distribution

- **Executive summary:** Henry.
- **Introduction:** Merrick.
- **Problem statement:** Shannon and Kamilia.
- **Service environment:** Omo.
- **Interest holders:** Abdulrahman.
- **Detailed requirements:**
 - **Functions:** Merrick.
 - **Objectives:** Kamilia.
 - **Constraints:** Shannon.
- **Conclusion:** Abdulrahman.

Gap, Need, and Scope Notes

- **Gap:** lack of means to identify first response in Skule Kup events.
- **Need:** a method of first-response detection among a variable group of contestants.
- **Scope:** the Pit / Skule Kup events, with Discipline Feud used as the main reference game context.





2026-02-04: Third Tutorial

- Debriefed **Client Meeting #1**.
- Worked on the **Project Requirements**.

2026-02-10: Third Team Meeting - Working on PR and Reviewing the PR

- Meeting focused on working on the **PR** and reviewing the PR draft.
- Talked about the **appendix** and what should be inside.
- Set individuals **internal deadlines** for the PR work.
- For future meeting, the regular team meeting start time was changed from **6:00 PM to 5:00 PM**.
- The **Project Requirements (PR)** due date was recorded as **February 15 at 1:00 PM**, and the internal submission time was set for **Saturday night**.
- Need to **measure the Pit** so the design and requirements could be made more precise.
- **Omo and Merrick** will visit and measure the Pit during the tutorial time.

2026-02-11: Fourth Tutorial

- Our team Continued work on the **PR**.
- The due date for the PR was recorded as **February 15 at 1:00 PM**.
- Thus, we set an internal deadline for the PR of **February 14 at 11:59 PM**.

2026-02-14 to 2026-02-22: Reading Week

- Reading Week.

2026-02-24: Fourth Team Meeting - Brainstorming Idea Generation

- Team meeting #4 focused on brainstorming and idea generation for the **CDS**.
- Talked about the **second client meeting** and identified possible meeting times.
- Started idea generation -> **magic solutions, blue-sky thinking, solo brainstorming, and group brainstorming**.
- The meeting also started narrowing ideas and making decisions for the **CDS**.

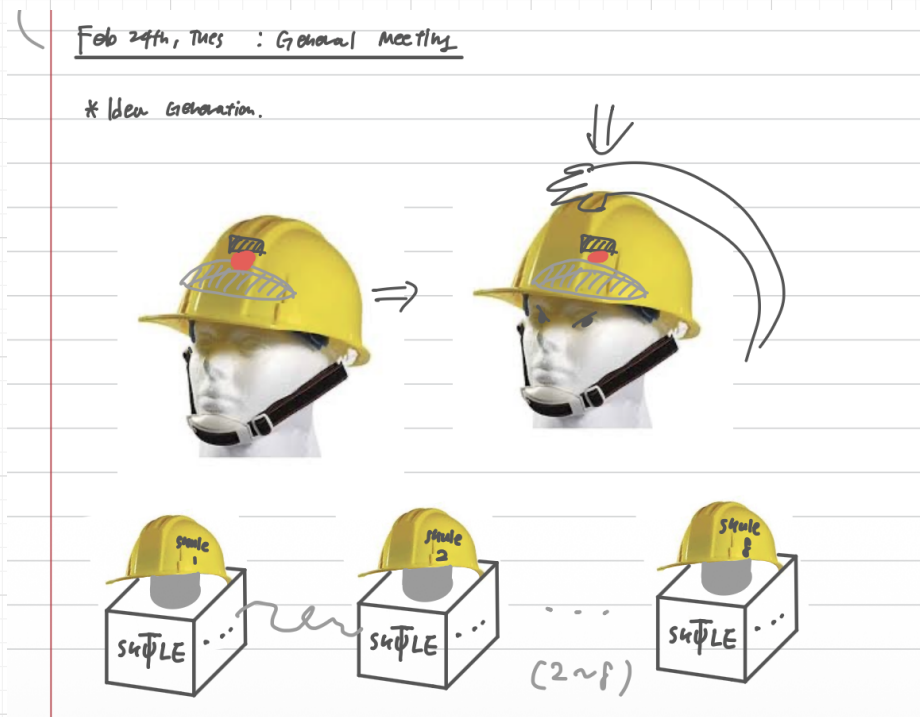


Ideas

- Handheld torch buzzers (combine with Henry's first idea).
 - Includes a button.
 - Henry: can be combined with a gyroscope sensor to detect motion.
- Gloves with pressure sensors - Shannon.
- Mobile app with button interface (Kahoot style) - Shannon.
- Hardhat buzzer / hardhat motion-sensor concept - Henry.
- Gun that shoots confetti when a participant wants to answer - Merrick.
- Foot pedal - Omo.
- Detect the first voice that says "buzz" - Abdulrahman.
- Buzzer connected to the host's computer that gives the names in order of who pressed first - Kamilia.

Personal Concept Sketch

- I generated a sketch of my idea during this meeting was a **Hardhat buzzer** concept.





2026-02-25: Fifth Tutorial - PR Debrief

- **Project Requirements** review and feedback from the **TA** and **EM**.
- Our team PR mostly received **below** and **meets** on the rubric, So the document needed more revision.
- A major feedback was **justification**. Some writing needed to be more general and explained more clearly.
- The big mistake was missing **living things**, user demographics, and assumptions for metric checks.

Debrief Note Summary – After discussing with TA and EM

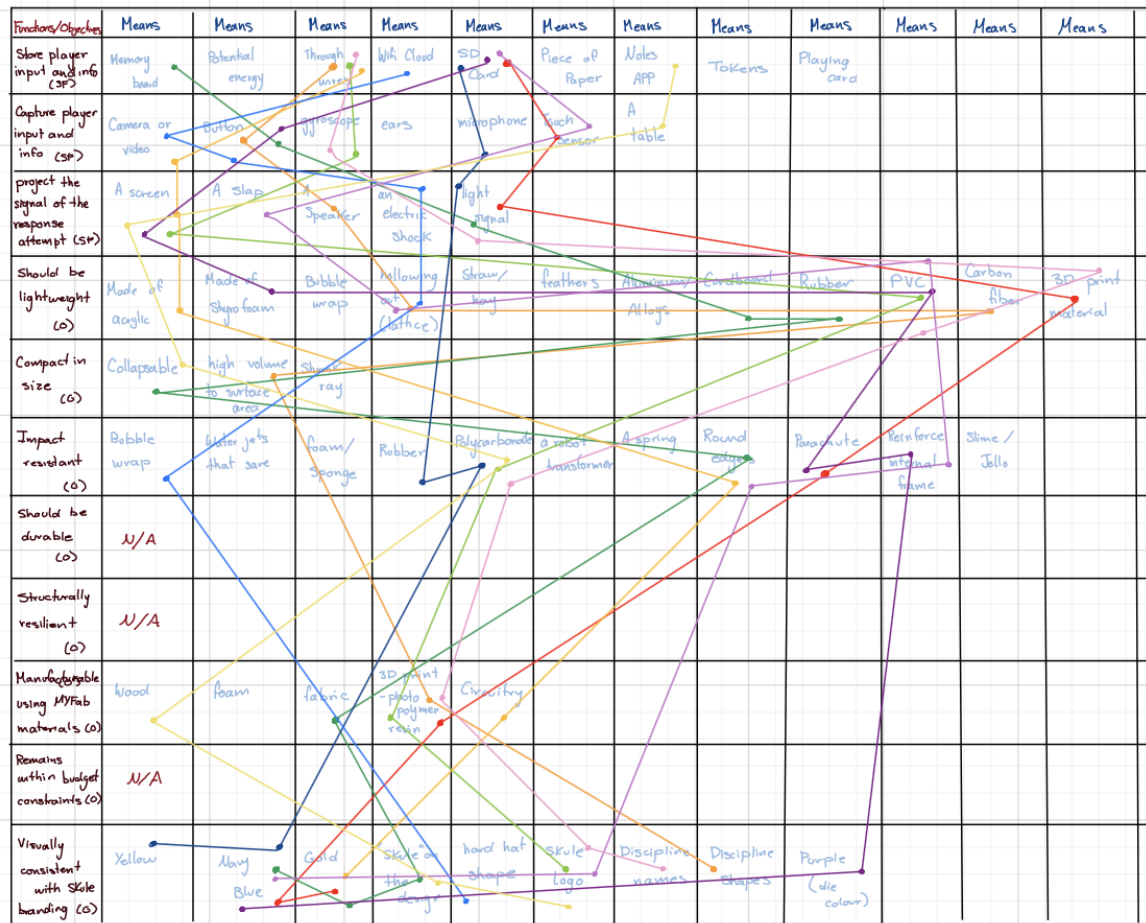
- **Executive summary:** start numbering at 1.0, describe the event context more clearly, and introduce the client earlier.
- **Introduction:** use more general wording.
- **Problem statement:** revise wording in several sentences.
- **Service environment:** add living things
- **Objectives:** add more justification in the actual document.
- **Constraints:** improve wording in certain areas.
- **General feedback:** citations needed work to better match IEEE style, and some sections needed more detailed introductions.

2026-03-03: Fifth Team Meeting - PR Revision and Morph-Chart Preparation

- Continued **idea generation** for CDS
- Finish **revising the PR**.
- Made the agenda for the upcoming **second client meeting**.
- The second client meeting for **March 4, 2026 at 3:00 PM** in **Galbraith Lobby**.

Morph Chart Work

- We made a **morph chart** to help generate better and more structured ideas.
- **Kamilia** led the morph-chart drafting during this stage.
- The chart helped us compare different functions, means, and combinations more systematically.



2026-03-04: Tutorial

- **Henry(ME)**, **Kamilia**, and **Shannon** looked into which materials would be needed to start building the prototype.
- Focused the project into more concrete **prototype planning and procurement**.

2026-03-04: Second Client Meeting - Galbraith Lobby @ 3PM

- The client had a **positive first reaction** to the PR.
- The client thought it was a good idea to use the **Arena rendering** as a reference.



PR Review with Client

- **Service environment:** replace the earlier “living things” note with **small birds** around the Pit in winter; cell service at the Pit was also noted as **not very strong**.
- **Interest holders:** the client felt the listed interest holders were accurate, preferred keeping the **host** and **director** separate, and noted that **participants** mattered more than U of T facilities in this context.
- **Functions:** the client understood and liked the listed functions.
- **Objectives:** the client liked the focus on **lightweight design** and **compactness**, agreed with most of the order, suggested switching **structural resilience** and **durability**, and said the durability target could be shortened to about **10k cycles**.
- **Constraints:** the client had no major issue with the listed constraints and said that **safety should rank above functionality** if trade-offs are needed.

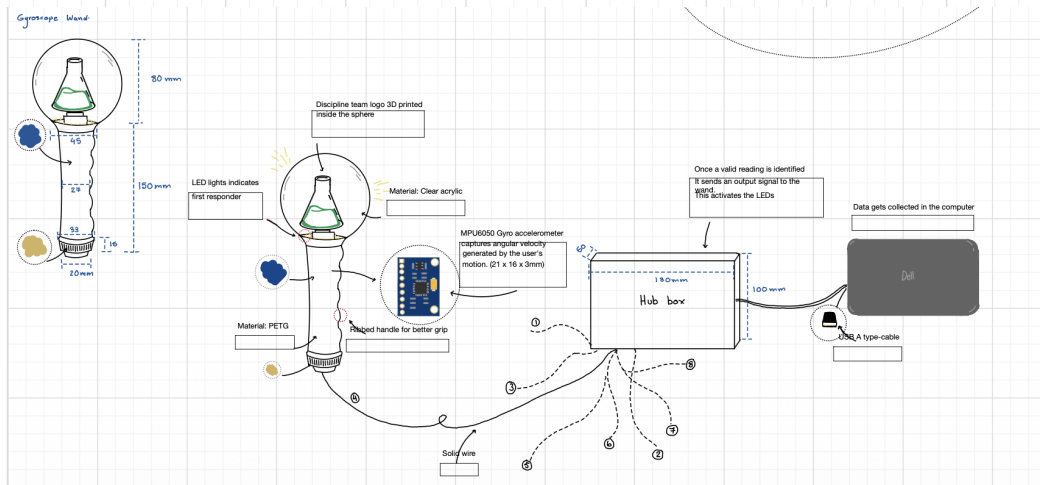
Idea Selection Feedback

- The client liked the **gyroscope** idea and the **hardhat** idea, and said both felt very Skule.
- The client preferred a **physical buzzer system** over an app.
- The client did **not** like the **confetti** idea because of cleanup.
- The client seemed to prefer ideas that were more **unconventional** and more **upper-body focused** (hands / arms) rather than feet.
- A smaller objective mentioned during the meeting was that the final design should be **fun**.

2026-03-21: Prototype Modeling using Blender Software

- I used **Blender** to model our prototype in a more realistic and professional way.
- I used **Kamilia's hand-drawn sketch** as the main reference while building the model.

Reference Sketch



Blender Model



2026-03-24: Eighth Team Meeting

- Allocated the remaining team tasks for prototype completion, testing, data work, and presentation preparation.
- We also made a timeline for finishing the **MoS** work before the final presentation.



Task Allocation

- **Procuring materials:** Shannon.
- **3D printing:** Kamilia and Henry.
- **Coding the design:** Henry.
- **Testing:** Omo and Abdulrahman.
- **Writing data:** Omo and Abdulrahman.
- **Data analysis:** Omo and Abdulrahman.
- **Writing the presentation:** Merrick.

Final Presentation Notes

- Think about the worst questions we could get and place those in the **appendix slides**.
- First introduce the **problem** that we were solving.
- Then introduce the **design**.

Timeline for Completing MoS

- Order materials from **MyFab** and **Amazon** – Kamilia.
- Go to MyFab, check the prints, and pick up materials – whole team – **11:30 AM, April 1**.
- Have the wand printed and assembled – Kamilia, Henry, and Shannon – **Friday, April 3**.
- Run the drop test – Abdulrahman and Omo – **Saturday, April 4**.
- Procure data and get it analyzed – Abdulrahman and Omo – **Sunday, April 5**.
- Finish the presentation – Merrick, with help from the rest of the team – **Sunday, April 5**.
- Practice the presentation – whole team – **Monday to Wednesday**.